

Department of Physical Science
Faculty of Applied Science
CSH4216 Research Project
Undergraduate Research Synopsis Submission Form

Student Information

Name	P.C.M Dharmasiri
Reg. No	2021/ASP/24
Degree Program	BSc (Hons) in Computer Science
Email	itsmechathura@outlook.com
Phone No	+94 717480048

Research Details

Proposed Research Title	Inferring Battery State-of-Charge (S and Predicting Overtaking Probability in Formula 1 Racing Using Machine Learning Under 2026 Power Unit Regulations
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Brief Description of the Research (100-150 words):

Problem Context

- F1's 2026 technical regulations introduce power units where electrical energy contributes ~50% of total power output (50:50 ICE-to-electric ratio)
- This makes battery State-of-Charge (SoC) management central to overtaking opportunity
- Team SoC data is proprietary , not publicly available for research

Proposed Solution

- A physics-informed method to estimate SoC from public FastF1 telemetry (speed, throttle, brake etc.)
- Grounded in FIA 2026 MGU-K power output and battery capacity specifications

Feature Set & Model

- Estimated SoC combined with: inter-driver gap, tyre compound, tyre age, and acceleration
- Features feed into a machine learning model (XGBoost) trained to predict overtaking probability

System Architecture

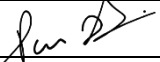
- Two-layer design:
 - Layer 1: Offline training pipeline built from historical 2026 race telemetry
 - Layer 2: Simulated real-time inference pipeline using a NoSQL time-series database (ScyllaDB)
- Reflects the practical constraints of genuine live F1 data access

Objectives of the Research:

1. **Estimate hidden battery data:** Build a method to calculate a Formula 1 car's Battery State-of-Charge (SoC) using telemetry data that's publicly available (throttle, brake, speed, etc.), based on the 2026 energy rules.
2. **Create useful features from the data:** Combine the estimated SoC with other race information gap to the car ahead, tyre compound, tyre age, and acceleration following the FIA's 2026 Power Unit regulations.
3. **Build a prediction model:** Train a machine learning model that uses these features to predict how likely a driver is to successfully overtake another car, based on past real race data.
4. **Build a two-part system:** Create one part that trains the model offline using past race data, and a second part that simulates real-time predictions using a NoSQL time-series database.
5. **Test and evaluate the model:** Check how well the trained model performs, and see if this approach could be practically useful for race strategy and F1 data analysis.

Student Declaration:

I hereby declare that the contents of this proposal synopsis are the result of my own independent work. No part of this document has been copied from any other source, and all references used have been duly acknowledged.

Student's Signature		Date	21/08/2026
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Supervisors Details

Primary Supervisor Name	Dr. S. Kirushanth		
Co-supervisors (Name & Institution/Address)	Mr. V. Athiththan (University of Vavuniya – Department of Physical Science)		
Primary Supervisors Approval	☐	Requires Revision	☐

Comments (if any):

Primary Supervisor Signature	
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Approval of Head

Approved	☐	Rejected	☐	Requires Revision	☐
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Comments:

Signature of Head of the Department		Date	
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